Interactive Prototype Outcome Evaluation 2b

Objective and validation metrics

For this evaluation I wanted to understand whether the changes I made after Evaluation 2a fixed the control problems, and whether moving the workspace onto a real table with passthrough made the Notion AR flow easier. In Evaluation 2a the concept held up but the controls did not. All three people tapped the grip instead of holding it, said the buttons felt random, and the first time VR user could not find the buttons. I was also checking whether the five design principles from the lectures still held up once the prototype was on a Quest 3S with passthrough.

IP2b changed the controls so that trigger is select and grip is grab, used only the right controller, added live button labels in view, used passthrough so the page sits on the real table instead of a virtual one, made the text slightly bigger, and ran on a Quest 3S instead of a Quest 2 for better passthrough quality.

My testing focused on five areas:

Testing aim	Success metric
Real table works as the workspace	Page placed on the real table without help.
New button layout makes sense	Steps done without me explaining the controls. Buttons do what people expect.
Moving the house works first time	House moved by holding grip without a hint (0 of 3 in Evaluation 2a).
Visible controller helps find buttons	New testers find the right button without help.
Concept is still easy and useful	Ease and usefulness from 1 to 5 (my observer ratings), compared with Evaluation 2 (4.1 and 3.9).

I counted it as a success if participants could finish the flow without me explaining the buttons, move the house without a hint, place the page on the real table alone and if my ease and usefulness ratings were at least as high as Evaluation 2.

I tested four participants on Thursday 22 October 2 in the Week 12 studio and the other 2 at home, on a Meta Quest 3S using only the right controller, with passthrough on and a real table. P1 and P2 had tested the earlier IP2a build in Evaluation 2. P3 was the tutor and new to Notion AR. P4 was also new. Before each person I had them face forward and press B to set the table and height. Before testing I briefly explained what the prototype is for and that the page goes on the table. I did not explain the controls unless someone was stuck. Each step was marked Yes if they did it alone, Mostly if they finished after a hint, or No if they could not. After each test I rated ease and usefulness from 1 to 5 based on how I saw them perform and they answered five short questions. I also used the think out loud protocol for capturing any additional notes. I did not use a standard questionnaire such as the System Usability Scale.

Results

Four participants completed the same flow: open the page list, select the House Sketch, place it on the real table, reveal the 2D sketch, lift it into a 3D house, and move the house around. The comments below are paraphrased from what they said. The raw notes are in Appendix A.

Step	P1 (returning)	P2 (returning)	P3 (tutor, new)	P4 (new)
1. Open pages	Yes	Yes	Yes	Yes
2. Place on real table	Yes	Yes	Mostly	Mostly
3. Reveal sketch	Yes	Yes	Yes	Yes
4. Lift into 3D	Yes	Yes	Yes	Yes
5. Move house	Yes	Yes	Yes	Yes
Ease (1–5, my rating)	4.9	4.8	4.5	4.3
Usefulness (1–5, my rating)	4.9	4.9	4.8	4.7

Overall, all four participants finished the flow and nobody failed a step. All four opened the page picker with A and could read all the text. All four used Trigger to reveal and lift, and all four moved the house by holding Grip without a hint. In Evaluation 2a nobody did that. P1 and P2 flew through, often without me saying anything, and finished every step alone. P3 and P4 both got stuck on placing the page. P3 put the controller on the table but kept pressing while it was flat and revealed the sketch early. P4 put the controller down flat and left it instead of pressing and had the wrong height. I stepped in for both. Ease averaged 4.63/5 (4.9, 4.8, 4.5 and 4.3). Usefulness averaged 4.83/5 (4.9, 4.9, 4.8 and 4.7). Both are up from Evaluation 2a, where ease averaged 4.1 and usefulness 3.9. That comparison needs caution though, because in Evaluation 2 the participants rated themselves after each test, and in this round I rated them from what I saw. The returning testers scored highest. The buttons did what people expected. The most confusing part was guessing how to place the page on the table and what they would improve is a visual instruction for that. The real table was a bit more of a challenge than the virtual one, because real tables can be messy, but people still placed the item once they knew it belonged there.

Analysis and insights

The results show that the control changes from Evaluation 2a worked. In Evaluation 2a everyone tapped the grip and said the buttons felt random. In this round every tester, including the two new ones, moved the house by holding Grip without a hint and the buttons performed as expected. That matches what I thought after Evaluation 2a. The problem was the mapping, not the people.

A key pattern was that once the buttons were clear, the hardest step moved to placing the page on the real table. Only the two new testers struggled there. Both got the controller onto the table but did not know when to press, or pressed while it was flat. Once I stepped in, they finished. The interaction works, but it is not discoverable on its own. This is the same pattern as the grip in Evaluation 2a. The first time a step works differently from the rest, people need a cue.

Another important pattern was that passthrough was worth the change from the virtual table. Everyone stayed on the real table and nobody switched to the virtual one. The Quest 3S made the AR clearer and better quality than the Quest 2 used in Evaluation 2. The real table was a bit harder to identify than the virtual one, but people still understood it was the workspace once they knew the page belonged there.

My ease and usefulness ratings both rose from Evaluation 2a, which fits the fixed controls and the clearer headset. The returning testers scored highest, so some of the gain may be learning from the earlier test. The two new testers still scored above Evaluation 2's averages, which suggests the design itself also improved, with the caution that the rating methods differ. The design was useful, but people sometimes missed what to do at the table step.

Design principle	Verdict	Evidence from IP2b
Text readability	Met	Everyone could read all the text.
Ergonomics	Met	Nobody stretched, bent or turned awkwardly. Nobody adjusted the table height with the thumbstick.
Brightness changes	Met	Everyone used passthrough the whole time. Nobody switched to the virtual table.
Avoiding simulator sickness	Met	Nobody reported dizziness or eye strain.
Button placement	Met	People pressed the right buttons. All four held Grip to move without a hint.

Evaluation of aims

The real table aim was partly validated. Everyone stayed on passthrough, but the two new testers needed me to step in for placement and I had already told everyone the page goes on the table, so this aim was not tested fully unaided.

The button layout aim was validated. Every tester used the buttons as expected and all four moved the house without a hint, which is the biggest change from Evaluation 2a.

The moving the house aim was validated. Nobody moved the house without a hint in Evaluation 2. In this round all four did.

The visible controller instructions was validated with some caution. The new testers found the right buttons without help once the flow started, but only two were new and I had already briefed them on the purpose.

The ease and usefulness aim was mostly validated. My ratings rose from 4.1 to 4.63 for ease and from 3.9 to 4.83 for usefulness, with the caution that the returning testers scored highest, so learning from Evaluation 2a may be part of that rise, and that Evaluation 2 used participant self-ratings while these scores are my observer ratings.

Concept iteration

Based on the evaluation, if this project were to continue i would keep the new controls and the passthrough table and change the table placement so the prototype matches what people already tried to do.

The next prototype should include:

1. A visual guide for placing the controller and page on the real table, including when to press while the controller is flat. This comes from P3 and P4 getting stuck there and from the shared request for visual instructions.
2. Keep the button layout. Trigger to select, place and lift, grip to grab and move, right hand only, with the visible controller and live labels. All four held grip to move without a hint.
3. Keep passthrough on the real table. Everyone stayed on it, nobody reported discomfort, and the Quest 3S made the AR clearer than the Quest 2.
4. Keep the slightly bigger text. Everyone could read all labels.

5. Test with more new people who have not seen the prototype, without briefing them that the page goes on the table, so the placement cue can be checked properly.

These changes are backed up with the testing results. I am keeping the parts that worked and changing the table placement that did not. Notion AR stays a workspace where a page's 2D sketch is lifted into a 3D object, now on the user's real table, with normal Quest controls. The difference is that table placement should explain itself the way the rest of the flow already does.

Reflection and future testing

This evaluation helped me understand that the changes from Evaluation 2a were the right ones. I swapped the buttons based on what three people tried to do, and four people, including two new ones, moved the house without a hint. The Quest 3S also helped. Passthrough was clearer and better quality than on the Quest 2. Moving from a fake table to passthrough was worth the extra work.

The method worked. Using the same flow, Yes or Mostly per step and the same 1 to 5 scales let me compare with Evaluation 2 and see that the problem had moved from the grip to the table step. I rated ease and usefulness based on how each person performed.

What did not work as well was the sample and some of my own process. I tested four people, not five, and the tutor was one of them, so only two were new. The ease and usefulness scores are observer ratings by me, not self-reported by the participants, so they carry my bias and are harder to compare with Evaluation 2a, where participants rated themselves. I also briefed people on the purpose and told them the page goes on the table, and I stepped in for P3 and P4 on placement, so I could no longer see whether they would have worked it out themselves. The returning testers scored highest, so a learning effect is a real caution on the rise in ratings.

For the future I want at least five new people with the visual table placement guide added and no briefing that the page goes on the table. I would also have participants rate ease and usefulness themselves, or use a standard questionnaire such as the System Usability Scale, and test more than one page. That will tell me whether the placement cue is enough for someone who has not seen the prototype before.

Appendix A: Raw participant notes

Participant 1

- Returning from IP2a.
- Flew through the flow, often without me saying anything. All steps Yes.
- Opened picker with A, placed on the real table alone, revealed and lifted with Trigger, moved the house by holding Grip without a hint.
- Ease of use (my rating): 4.9/5. Usefulness (my rating): 4.9/5.

Participant 2

- Returning from IP2a.
- Flew through. All steps Yes.
- Opened picker with A, placed on the real table alone, revealed and lifted with Trigger, moved the house by holding Grip without a hint.
- Ease of use (my rating): 4.8/5. Usefulness (my rating): 4.9/5.

Participant 3

- Tutor. New to Notion AR.
- Open, Reveal, Lift and Move: Yes. Place: Mostly.
- Placed the controller on the table but kept pressing buttons while it was flat and revealed the sketch early. Height and controller-on-table placement were unclear. I stepped in.
- Moved the house by holding Grip without a hint.
- Ease of use (my rating): 4.5/5. Usefulness (my rating): 4.8/5.
- Most confusing: guessing how to place the page on the table.

Participant 4

- New to Notion AR.
- Open, Reveal, Lift and Move: Yes. Place: Mostly.
- Put the controller down flat and left it instead of pressing. Wrong height. I stepped in.
- Moved the house by holding Grip without a hint.
- Ease of use (my rating): 4.3/5. Usefulness (my rating): 4.7/5.
- Most confusing: guessing how to place the page on the table.